ORGANIC CHEMISTRY · ASYNCHRONOUS ACTIVITY 03

# Fire Up the Locants

NAME IT, DRAW IT

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**BSChE-2A**



**Charizard says hi**



# PART A -

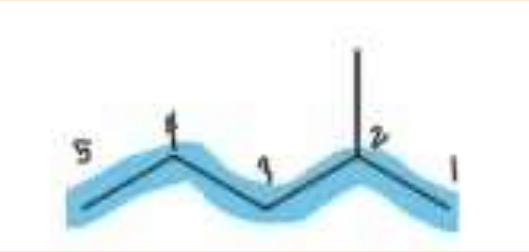
## STRUCTURE → NAME

# CARD A01-Foundation

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C6H14

## 1 NUMBERED PARENT



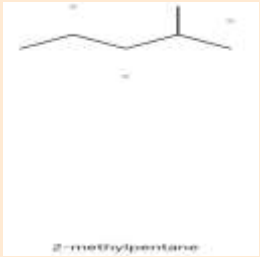

## 2 NAMING TRAIL

- **Parent**: longest chain = 5 C → pentane.
- **Numbering**: left → right gives 4; right → left gives 2. 2 < 4, so number from the right.
- **Substituents**: methyl on C2 → 2-methyl.

## 3 MY NAME

*2-methylpentane*

## 4 THE ALKANE CHECK



- **Returned Name**: 2-methylpentane

✓ Match

The rebuilt structure is a 5-C chain with one CH3 on the second carbon. The tool applies the same rules as the naming trail: longest chain first, then lowest locant (2 beats 4). Both therefore give the same name.

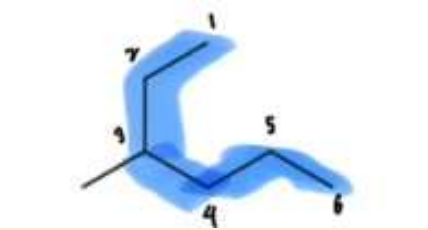
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A02-Foundation

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C7H16

## 1 NUMBERED PARENT



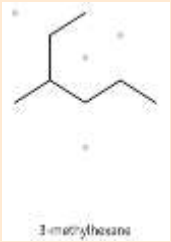

## 2 NAMING TRAIL

- **Parent**: longest chain = 6 C → hexane. The horizontal path is only 5 C.
- **Numbering**: top → bottom gives 3; the other direction gives 4. 3 < 4, so number from the top.
- **Substituents**: the short left stub is a methyl on C3 → 3-methyl

## 3 MY NAME

*3-methylhexane*

## 4 THE ALKANE CHECK



- **Returned Name**: 3-methylhexane

✓ Match

The structure has a 6-C path through the branch point, with one methyl on the third carbon. The tool traces the longest chain through branch points, so hexane is chosen over the 5-C horizontal path, and 3 is the lowest locant

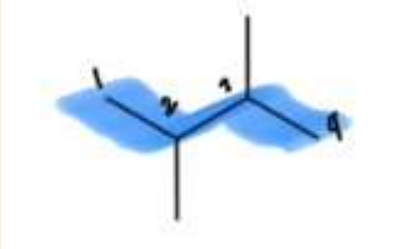
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A03-Foundation

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C6H14

## 1 NUMBERED PARENT



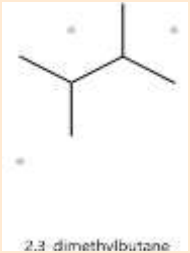

## 2 NAMING TRAIL

- **Parent**: longest chain = 4 C → butane.
- **Numbering**: both directions give 2,3, so the sets tie and either direction is fine.
- **Substituents**: methyl on C2 and methyl on C3 → 2,3-dimethyl

## 3 MY NAME

*2,3-dimethylbutane*

## 4 THE ALKANE CHECK




- **Returned Name**: 2,3-dimethylbutane
  ✓ Match
- The structure is a 4-C chain with one methyl on each middle carbon. The locant set is {2,3} from both ends, and the tool writes identical groups with the prefix di- and a comma between locants.

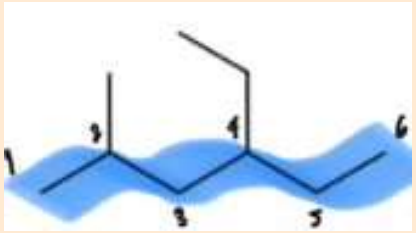
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A04-Developing

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C9H20

## 1 NUMBERED PARENT



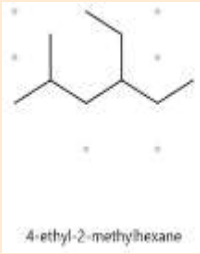
## 2 NAMING TRAIL

- **Parent**: 6 C → hexane. Another 6-C path runs through the ethyl, but it carries the same two substituents (a methyl and an ethyl), so it leads to the same name.
- **Numbering**: left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.
- **Substituents**: ethyl on C4, then methyl on C2 → 4-ethyl-2-methyl. Alphabetical order sets the citation order, not the locants.

## 3 MY NAME

*4-ethyl-2-methylhexane*

## 4 THE ALKANE CHECK




- **Returned Name**: 4-ethyl-2-methylhexane

✓ Match

The tool numbers from the end that gives {2,4}, not {3,5}. It cites ethyl before methyl because e comes before m alphabetically, even though ethyl has the higher locant.

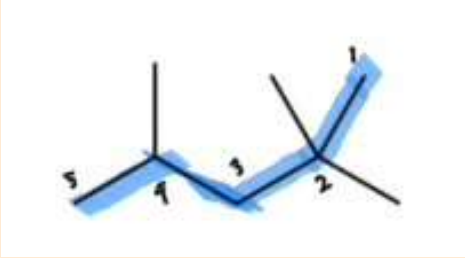
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A05-Developing

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C8H18

## 1 NUMBERED PARENT



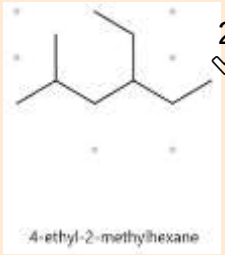

## 2 NAMING TRAIL

- **Parent**: longest chain = 5 C → pentane.
- **Numbering**: left → right gives 2,4,4; right → left gives 2,2,4. The first terms tie at 2. At the second term 2 < 4, so number from the right.
- **Substituents**: methyl on C2, C2 and C4 → 2,2,4-trimethyl.

## 3 MY NAME

*2,2,4-trimethylpentane*

## 4 THE ALKANE CHECK




- **Returned Name**: 2,3-dimethylbutane

✓ Match

The tool numbers from the end that gives {2,4}, not {3,5}. It cites ethyl before methyl because e comes before m alphabetically, even though ethyl has the higher locant.

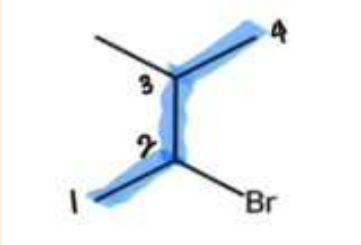
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A06-Developing

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C5H11Br

## 1 NUMBERED PARENT



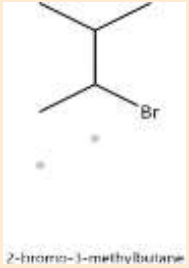

## 2 NAMING TRAIL

- **Parent**: longest chain = 4 C → butane.
- **Numbering**: both directions give {2,3}, so the locant sets tie.
- **Tie-break**: bromo (b) precedes methyl (m) alphabetically, so bromo gets C2.
- **Substituents**: bromo on C2, methyl on C3 → 2-bromo-3-methyl.

## 3 MY NAME

*2-bromo-3-methylbutane*

## 4 THE ALKANE CHECK




- **Returned Name**: 2-bromo-3-methylhexane

✓ Match

The locants tie at {2,3}, so the tool applies the alphabetical tie-break and gives the lower number to bromo. The halogen is written as a prefix (bromo) and is not part of the parent name.

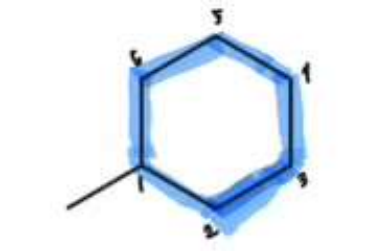
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A07-Developing

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C7H14

## 1 NUMBERED PARENT



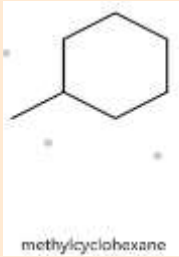
## 2 NAMING TRAIL

- **Parent**: the ring is the parent, 6 C → cyclohexane.
- **Numbering**: there is only one substituent, so there are no locants to compare. The substituent carbon is C1, and no locant is written.
- **Substituents**: methyl.

## 3 MY NAME

*methylcyclohexane*

## 4 THE ALKANE CHECK




- **Returned Name**: methylcyclohexane

✓ Match

The structure is a 6-C ring with one CH3. The tool takes the ring as the parent and, because there is only one substituent, writes no locant (the name is not 1-methylcyclohexane).

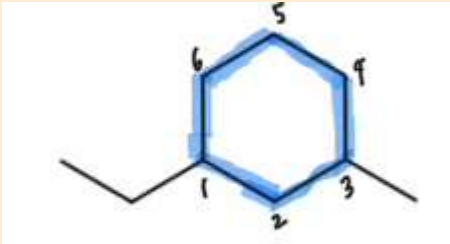
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A08-Challenge

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C8

## 1 NUMBERED PARENT



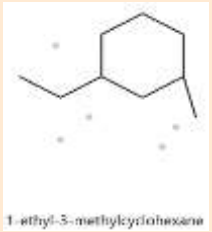

## 2 NAMING TRAIL

- **Parent**: the ring is the parent → cyclohexane.
- **Numbering**: going the short way round gives {1,3} whichever substituent is C1. The long way round gives {1,5}. {1,3} wins, and the two short-way sets tie.
- **Tie-break**: ethyl (e) precedes methyl (m), so ethyl gets C1.
- **Substituents:** ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

## 3 MY NAME

*1-ethyl-3-methylcyclohexane*

## 4 THE ALKANE CHECK




- **Returned Name**: 1-ethyl-3-methylcyclohexane

✓ Match

The two substituents are separated by one ring carbon, so the lowest locant set is {1,3}. The sets tie in both directions, so the alphabetical rule places ethyl on C1.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A09-Challenge

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C10H22

## 1 NUMBERED PARENT



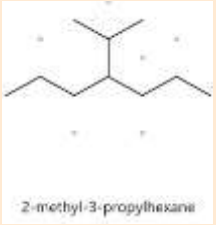

## 2 NAMING TRAIL

- **Parent**: Parent (7 C) highlighted; the 3-C branch is a substituent.
- **Numbering**: both directions give 4, so the sets tie.
- **Substituents**: a 3-carbon branch attached through its middle carbon is isopropyl, also written (1-methylethyl) or propan-2-yl, on C4.

## 3 MY NAME

*4-isopropylheptane*

## 4 THE ALKANE CHECK




- **Returned Name**: 4-isopropylhexane

✕ Did Not Match

The tool numbers from the end that gives {2,4}, not {3,5}. It cites ethyl before methyl because e comes before m alphabetically, even though ethyl has the higher locant.

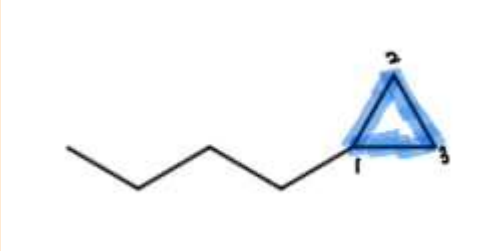
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# CARD A10-Challenge

PART A - STRUCTURE → NAME

MOLECULAR FORMULA: C7H14

## 1 NUMBERED PARENT



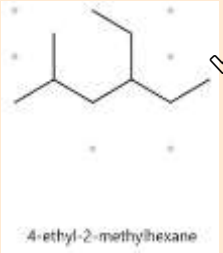

## 2 NAMING TRAIL

- **Parent**: the ring is the parent under this activity's rule, even though the chain (4 C) is longer than the ring (3 C) → cyclopropane.
- **Numbering**: one substituent, so no locant is needed.
- **Substituents**: unbranched 4-carbon chain → butyl.

## 3 MY NAME

*butylcyclopropane*

## 4 THE ALKANE CHECK




- **Returned Name**: butylcyclopropane

✓ Match

The tool follows the same ring-as-parent rule as this activity. It names the 3-C ring as the parent and the chain as butyl, with no locant for a single substituent.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

# PART B -

## NAME → STRUCTURE

# CARD B01-Foundation

PART B - NAME → STRUCTURE

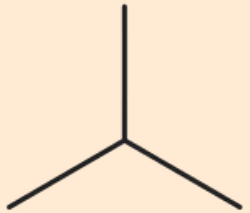
Given Name: 2-methylpropane

## 1 NAME DECODED

- Parent: propane → a 3-carbon chain.
- Substituents: methyl on C2.
- Expected formula: 3 + 1 = 4 C;

open chain → $C\square H_2\square_{+2}$ = $C_4H_{10}$

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_4H_{10}$
- **Open chain:** $C\square H_2\square_{+2}$ = $C_4H_{10}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 2-methylpropane
- ✓ **Same as the given name**
- The longest chain is 3 C (propane) with one methyl on the middle carbon, so the locant is 2 from either end.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B02-Foundation

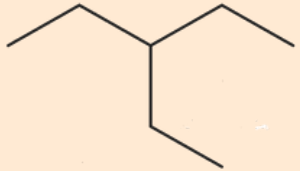
PART B - NAME → STRUCTURE

Given Name: 3-ethylpentane

## 1 NAME DECODED

- Parent: pentane (5-carbon chain)
- Substituents: ethyl on C3.
- Expected formula: 5 + 2 = 7 C; open chain → $C_7H_{16}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- Counted from the drawing: $C_7H_{16}$
- Open chain: $C\square H_{2}\square_{+2}$ = $C_7H_{16}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 2-methylpropane
- ✓ **Same as the given name**
- The longest chain is 3 C (propane) with one methyl on the middle carbon, so the locant is 2 from either end.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B03-Foundation

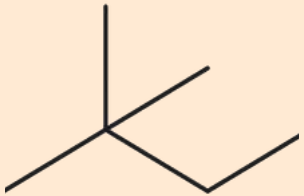
PART B - NAME → STRUCTURE

Given Name: **2,2-dimethylbutane**

## 1 NAME DECODED

- Parent: butane (4-carbon chain)
- Substituents: two methyls, both on C2.
- Expected formula: 4 + 2 = 6 C; open chain → $C_6H_{14}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_6H_{14}$.
- **Open chain:** $C\square H_{2}\square_{+2}$ = $C_6H_{14}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 2,2-dimethylbutane
- ✓ **Same as the given name**
- The longest chain is 4 C (butane). Numbering gives 2,2 from one end and 3,3 from the other, so 2,2 wins.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B04-Developing

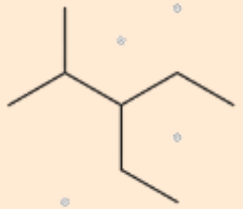
PART B - NAME → STRUCTURE

Given Name: **3-ethyl-2-methylpentane**

## 1 NAME DECODED

- Parent: pentane (5-carbon chain)
- Substituents: methyl on C2; ethyl on C3. Ethyl is cited first because e comes before m, even though its locant is higher.
- Expected formula: 5 + 1 + 2 = 8 C; open chain → $C_8H_{18}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_8H_{18}$.
- **Open chain:** $C\square H_{2}\square_{+2}$ = $C_8H_{18}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 3-ethyl-2-methylpentane
- ✓ **Same as the given name**
- Numbering gives {2,3} from one end and {3,4} from the other, so {2,3} wins. Ethyl is cited before methyl alphabetically.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B05-Developing

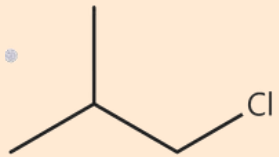
PART B - NAME → STRUCTURE

Given Name: **1-chloro-2-methylpropane**

## 1 NAME DECODED

- Parent: propane (3-carbon chain)
- Substituents: chloro on C1; methyl on C2.
- Expected formula: 3 + 1 = 4 C; open chain → $C_4H_{10}$; Cl replaces one H → $C_4H_9Cl$.

## 2 BOND LINE STRUCTURE




## 3 FORMULA

- **Counted from the drawing:** $C_4H_9Cl$.
- **Open chain:** $C\square H_2\square_{+2}$ = $C_4H_{10}$; minus one H for Cl = $C_4H_9Cl$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 1-chloro-2-methylpropane
- ✓ **Same as the given name**
- Numbering gives {1,2} from the Cl end and {2,3} from the other, so {1,2} wins. Chloro is cited before methyl alphabetically.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B06-Developing

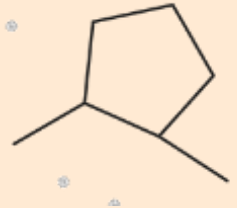
PART B - NAME → STRUCTURE

Given Name: **1,2-dimethylcyclopentane**

## 1 NAME DECODED

- Parent: cyclopentane (5-carbon ring)
- Substituents: methyl on C1; methyl on C2.
- Expected formula: 5 + 2 = 7 C; one ring → $C_7H_{14}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_7H_{14}$.
- **One ring:** C□$H_2$□ = $C_7H_{14}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 1,2-dimethylcyclopentane
- ✓ **Same as the given name**
- The ring is the parent. Numbering the short way round gives {1,2}, and the long way gives {1,5}, so {1,2} wins.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B07-Developing

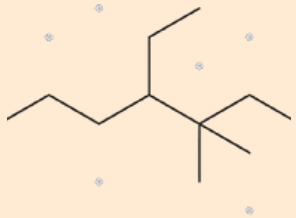
PART B - NAME → STRUCTURE

Given Name: **4-ethyl-3,3-dimethylheptane**

## 1 NAME DECODED

- Parent: heptane (7-carbon chain)
- Substituents: two methyls on C3 (3,3-dimethyl); ethyl on C4. Ethyl is cited first because e comes before m.
- Expected formula: 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_{11}H_{24}$.
- **Open chain:** $C\square H_{2}\square_{+2}$ = $C_{11}H_{24}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 4-ethyl-3,3-dimethylheptane
- ✓ **Same as the given name**
- The longest chain is 7 C. Numbering gives {3,3,4} from one end and {4,5,5} from the other, so {3,3,4} wins. Ethyl is cited before dimethyl.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B08-Challenge

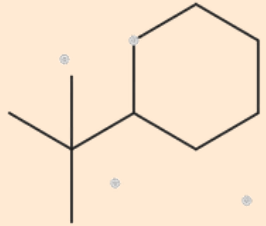
PART B - NAME → STRUCTURE

Given Name: **tert-butylcyclohexane**

## 1 NAME DECODED

- Parent: cyclohexane (6-carbon ring)
- Substituents: tert-butyl, $(CH_3)_3C–$, a 4-carbon group, on C1. A single substituent on a ring needs no locant.
- Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_{10}H_{20}$.
- **One ring:** C□$H_2$□ = $C_{10}H_{20}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** (1,1-dimethylethyl)cyclohexane
- ✓ **Same as the given name**
- tert-Butyl and (1,1-dimethylethyl) are equivalent names for the same group (Appendix B.3).

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B09-Challenge

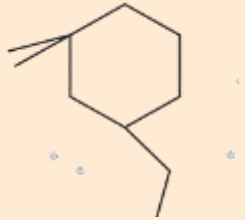
PART B - NAME → STRUCTURE

Given Name: **3-ethyl-1,1-dimethylcyclohexane**

## 1 NAME DECODED

- Parent: cyclohexane (6-carbon ring)
- Substituents: two methyls on C1 (1,1-dimethyl); ethyl on C3.
- Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_{10}H_{20}$.
- **One ring:** C□$H_2$□ = $C_{10}H_{20}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane:** 3-ethyl-1,1-dimethylcyclohexane
- ✓ **Same as the given name**
- Numbering gives {1,1,3} one way round the ring and {1,1,5} the other, so {1,1,3} wins. Ethyl is cited before dimethyl.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

# CARD B10-Challenge

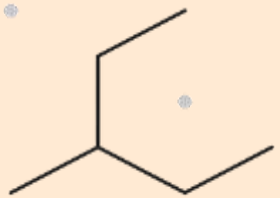
PART B - NAME → STRUCTURE

Given Name: **2-ethylbutane**

## 1 NAME DECODED

- Parent: butane (4-carbon chain)
- Substituents: ethyl on C2.
- Expected formula: 4 + 2 = 6 C; open chain → $C_6H_{14}$.

## 2 BOND LINE STRUCTURE



## 3 FORMULA

- **Counted from the drawing:** $C_6H_{14}$.
- **Open chain:** $C\square H_{2\square+2}$ = $C_6H_{14}$ ✓

## 4 BACK-CHECK

- **Returned by The Alkane**: 3-methylpentane
- ✗ Different from the given name
- Rule broken: the longest-chain rule (Appendix B.1, step 1). The path ethyl → C2 → C3 → C4 has 5 carbons, which is longer than the 4-carbon "butane" chain. The parent is therefore pentane, and the leftover $CH_3$ is a methyl on C3 (position 3 from either end).

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded